

specify inputs(rule)

I am testing a loan approval program named Loan.py for discrimination.
Please identify the input parameters required for fairness testing.

For each input, provide:

1. Input name
2. Data type
3. Possible values or valid range
4. Whether it can be considered a protected attribute
5. Why it may or may not be relevant for discrimination testing

Assume the goal is to test whether the loan decision changes unfairly when protected attributes such as gender are changed while other attributes remain fixed.

generate testcase(rule)

Using the input specification for Loan.py, generate fairness test cases for discrimination testing.

For each test case:

1. Create a base applicant profile
2. Create a counterfactual profile by changing only one protected attribute
3. Keep all other attributes the same
4. Predict whether the output should be the same or different
5. Explain why this test case may reveal discriminatory behavior

Return the test cases in a python format.

result(rule)

Assume the generated test cases were executed on Loan.py.

Given the outputs for each base and counterfactual pair, calculate:

1. Total number of test pairs
2. Number of discriminatory pairs where the loan decision changed
3. Percentage of discriminatory samples
4. Whether the program should be considered discriminatory based on the observed percentage

Use this formula:

$$\text{Discriminatory Sample Percentage} = (\text{Number of discriminatory pairs} / \text{Total number of test pairs}) \times 100$$

Summarize the fairness testing results for Loan.py.

Include:

1. Total number of generated test cases
2. Total number of discriminatory samples found
3. Percentage of discriminatory samples
4. Time required to generate the tests
5. Time required to find the first discriminatory sample, if available

actionable next(rule)

Based on the discrimination testing results for Loan.py, suggest actionable next steps.

For each suggestion, explain:

1. What action should be taken
2. Why it is needed
3. How it can reduce or help investigate discrimination

specify inputs(ml)

I am testing a machine learning based Adult Income prediction system for discrimination.

The program predicts whether a person's income is greater than 50K.

Please identify:

1. Input parameters used by the model
2. Data type of each parameter
3. Valid value ranges or categories
4. Which attributes are protected attributes
5. Why these protected attributes are important for fairness testing

Focus especially on demographic attributes such as Sex.

generate testcase(ml)

Generate counterfactual fairness test cases for the Adult Income prediction system.

Use the following fairness testing strategy:

- Create a base input profile
- Create a counterfactual profile by changing only one protected attribute
- Keep all non protected attributes fixed
- Compare whether the model prediction changes

Protected attributes:

- Sex

Non protected attributes:

- Age
- Race
- Education_Num
- Hours_per_week

For each test pair:

1. Show the base profile
2. Show the counterfactual profile
3. Explain why this pair may reveal discrimination

Present the results in python file.

result(ml)

Assume the generated test cases are executed on the Adult Income prediction program.

For each counterfactual pair:

1. Execute the prediction
2. Compare outputs between the base and counterfactual inputs
3. Mark the pair as discriminatory if the prediction changes after modifying only the protected attribute

Use:

- 1 = income >50K
- 0 = income <=50K

Return:

- Total number of tested pairs
- Number of discriminatory pairs
- Percentage of discriminatory samples

Summarize the fairness testing results for the Adult Income ML model.

Include:

1. Total generated test cases
2. Number of discriminatory samples
3. Discrimination percentage
4. Total runtime required for test generation and execution
5. Time to first discriminatory sample

Present the results in an academic evaluation style.

actionable next(ml)

Analyze the fairness behavior of the Adult Income Random Forest model.

Discuss:

1. Whether the model appears sensitive to protected attributes
2. Possible reasons behind discriminatory predictions
3. Whether indirect discrimination through correlated attributes may exist
4. How class imbalance or dataset bias may influence fairness
5. Risks of using such predictions in real world applications

Based on the fairness testing results of the Adult Income ML model, suggest actionable next steps.

Include:

1. Fairness mitigation strategies
2. Suggestions for rerunning discrimination testing after mitigation